

Traversal-Based Relative Positional Encodings for Graph Neural Networks

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Abstract

Message-passing graph neural networks are bounded by the 1-Weisfeiler–Leman test, and positional encodings are the standard remedy. Traversal-based encodings are almost always *absolute*: a node is described by its distance to a few anchors, inheriting whatever arbitrariness chose them. We study the *relative* alternative, a function of a node *pair*. Since every edge of a sparse graph joins nodes at distance one, such an encoding is meaningful only on a rewired graph, so we attach the sinusoidally encoded hop distance $d(u, v)$ to every pair within r hops and inject it into GCN and GAT. We place the result exactly in the expressiveness hierarchy—strictly above 1-WL, no stronger than 3-WL, strictly strengthened by counting shortest paths—each claim confirmed on an explicit witness, the last found by exhaustive search and shown to be minimal. Expressive power is not enough: on a vertex-transitive graph with uniform features the normalised aggregation of both GCN and GAT algebraically cancels any edge weighting, so an encoding injected that way provably cannot help, and empirically scores exactly chance. Our pre-registered prediction that the encoding helps on sparse, long-diameter graphs failed in the opposite direction—our largest gain (+21.3 points) is on the dataset we predicted would be degenerate—and the architecture, not the graph, decides whether it helps at all.

1 Introduction

Message-passing graph neural networks (MPNNs) aggregate information over local neighbourhoods, and their ability to distinguish graphs is bounded by

the 1-Weisfeiler–Leman colour refinement test [Xu et al., 2019, Morris et al., 2019]. The bound is not a technicality: it means an MPNN cannot see global symmetry, cannot measure how far apart two nodes are, and cannot separate graphs that agree locally but differ globally.

Positional encodings (PEs) are the standard response. They enrich node features with a structural signal computed outside the message-passing recursion, and they come in two families. *Absolute* PEs assign a vector to each node—Laplacian eigenvectors [Dwivedi et al., 2023, Dwivedi and Bresson, 2021], random-walk return probabilities [Dwivedi et al., 2022], or distances to a set of anchors [You et al., 2019, Li et al., 2020]. *Relative* PEs assign a value to each node *pair*, and are the norm in graph transformers, where a pairwise bias is added to every attention logit [Ying et al., 2021, Rampášek et al., 2022].

Traversal-based encodings have so far lived almost entirely in the absolute family. This is natural—a breadth-first search from an anchor produces a per-node distance, which drops straight into the node feature vector—but it carries a cost that is easy to overlook. The encoding is a function of the anchor set, so two runs that pick different anchors describe the same graph differently, and the choice has to be made by some heuristic (highest degree, say) whose ties must then be broken arbitrarily. The relative formulation has no anchors and no such choice: $d(u, v)$ is a property of the pair.

The obstacle is that a relative encoding has nowhere to live in a sparse MPNN. Messages travel along edges, and every edge joins nodes at distance exactly one, so a function of $d(u, v)$ restricted to E is constant. Making the relative view meaningful

requires changing which pairs exchange messages—that is, rewiring [Brüel-Gabrielsson et al., 2023, Topping et al., 2022]. We adopt the natural construction: breadth-first search to depth r from every node, an edge for every pair within r hops, and the hop distance as that edge’s feature.

This coupling has a consequence we take seriously throughout. Rewiring is itself known to help, by shortening paths and relieving over-squashing [Alon and Yahav, 2021], so an improvement over a plain baseline does not by itself credit the positional signal. We therefore run the rewired graph with a *constant* edge feature as an explicit control, and read the main result only against it.

Our contributions:

- **A relative traversal encoding** (T-RPE) that is anchor-free and permutation invariant, costs one breadth-first traversal per node, and reaches GCN and GAT through two complementary channels (§3).
- **An exact position in the expressiveness hierarchy** (§4): strictly above 1-WL, no stronger than 3-WL, and strictly strengthened by shortest-path counts. Every claim is attached to an explicit witness pair, and the shortest-path witness is found by exhaustive search over all graphs on $n \leq 6$ nodes, establishing that no smaller one exists.
- **A negative result about injection** (Proposition 4): because GCN and GAT both *normalise* their aggregation, an edge-weight or attention-bias injection cancels algebraically on vertex-transitive graphs with uniform features. A relative encoding delivered that way cannot help on CSL no matter how expressive it is, and measurably does not. Prior work reports relative encodings almost exclusively inside dense attention, where the issue never surfaces.
- **An evaluation designed around the mechanism** (§5), contrasting a sparse long-diameter benchmark with a dense short-diameter one, and separating the contribution of the encoding from that of the rewiring it requires.

2 Related Work

Absolute positional encodings. Dwivedi et al. [2023] introduced Laplacian eigenvector PEs,

which place nodes in a global spectral coordinate system. They are expressive but costly— $\mathcal{O}(|V|^3)$ for a dense decomposition—and each eigenvector is defined only up to sign, an ambiguity usually handled by randomising the sign during training or by architectures invariant to it [Lim et al., 2023]. Random-walk PEs [Dwivedi et al., 2022] instead record the probability of returning to a node after k steps; they are cheaper and sign-free, but stochastic in origin. You et al. [2019] and Li et al. [2020] use distances to sampled or selected anchor nodes, which is the closest prior work to ours in signal but not in form: the encoding is per-node and anchor-dependent.

Relative positional encodings. In sequence models, relative position has long been preferred to absolute [Vaswani et al., 2017]. The graph analogue appears in graph transformers, where Graphormer [Ying et al., 2021] adds a learned bias indexed by shortest-path distance to every attention logit, and GraphGPS [Rampášek et al., 2022] unifies such biases with local message passing. These models pay $\mathcal{O}(|V|^2)$ for dense attention. Our question is whether the relative view survives being brought back to a *sparse* architecture, where the bias must be carried by explicitly added edges rather than by all-pairs attention.

Rewiring. Alon and Yahav [2021] identified over-squashing, and Topping et al. [2022] tied it to negatively curved edges, motivating rewiring schemes that add or reweight edges. Closest to us, Brüel-Gabrielsson et al. [2023] rewire a graph using positional information and feed the encoding as an edge attribute, showing gains across architectures. Our construction follows theirs; what we add is the traversal-derived signal, the expressiveness analysis, and—because the two are confounded by construction—a control that separates the effect of the rewiring from that of the encoding it carries.

Expressiveness. Xu et al. [2019] and Morris et al. [2019] established the 1-WL bound. Distance-aware refinements sit above it: Li et al. [2020] analyse distance encoding, and Zhang et al. [2023] define generalised-distance WL (GD-WL), showing that shortest-path distance already yields a test strictly stronger than 1-WL. We do not propose a new WL variant; we identify which known one

our encoding realises, and then bound it from both sides.

3 Method

Traversal. For a graph $G = (V, E)$ and a radius r , a breadth-first traversal from each $u \in V$ to depth r yields the hop distance $d(u, v)$ for every v within r hops, and—at no extra cost—the number $\#sp(u, v)$ of shortest paths realising it. Each traversal is $\mathcal{O}(|V| + |E|)$, so the encoding costs $\mathcal{O}(|V|(|V| + |E|))$ per graph and is computed once.

In practice we compute both quantities algebraically rather than with an explicit per-source traversal, which would mean $|V|$ interpreter-level loops per graph. Writing A for the adjacency matrix, $(A^k)_{uv}$ counts *walks* of length k , so the smallest k with $(A^k)_{uv} > 0$ is exactly $d(u, v)$; and at $k = d(u, v)$ every length- k walk is necessarily a shortest *path*, so $(A^{d(u, v)})_{uv} = \#sp(u, v)$. Both therefore fall out of r matrix products. The equivalence is verified against a reference breadth-first search.

Rewiring. We take $E_r = \{(u, v) : 1 \leq d(u, v) \leq r\}$. At $r = 1$ this is the original graph and the encoding below is constant, which makes $r = 1$ a built-in degenerate control rather than only a hyperparameter setting.

Encoding. Each rewired edge carries $\gamma(d(u, v)) \in \mathbb{R}^p$, a sinusoidal embedding of its hop distance in the sense of Vaswani et al. [2017], so that nearby distances receive nearby representations instead of arbitrary unrelated ones. A small MLP maps $\gamma(d)$ —optionally concatenated with $\log(1 + \#sp(u, v))$ —to the edge representation e_{uv} .

Injection. The encoding enters through two channels. The first is the natural one for each architecture: in GAT, e_{uv} is an additive bias on the attention logit of edge (u, v) , which is what a relative positional encoding means in the transformer sense; in GCN, which has no edge-feature channel, e_{uv} becomes a learned scalar gate $w_{uv} = \sigma(\mathbf{w}^\top e_{uv})$ used as an edge weight.

The second channel is not optional, for reasons Proposition 4 makes precise. Each node also re-

ceives its aggregated relative-distance profile,

$$p_v = \sum_{u: 1 \leq d(u, v) \leq r} \gamma(d(u, v)), \quad x_v \leftarrow [x_v \parallel \text{MLP}(p_v)],$$

which is anchor-free, permutation invariant, and exactly the aggregation GD-WL performs. For the $\#sp$ variant the summand is $[\gamma(d), \log(1 + \#sp) \cdot \gamma(d)]$, which retains distance and path count *jointly*; summing the two separately would discard which path counts belong to which distance, and that joint information is precisely what §5 relies on.

Invariance. Since $d(u, v)$ and $\#sp(u, v)$ are isomorphism invariants of the pair, e_{uv} is permutation equivariant and the pooled model output is permutation invariant. This is worth stating because the obvious alternative traversal signal does *not* have the property: a BFS *visit-order* index depends on the order in which neighbours happen to be enumerated, so relabelling the nodes changes the encoding and the model is not a function of the graph at all. We verify both the invariance of our encoding and the failure of the order-based alternative empirically.

4 Expressive Power

Write $c^t(v)$ for the colour of v after t refinement rounds. 1-WL refines by $c^{t+1}(v) = \text{HASH}(c^t(v), \{\{c^t(u) : u \in N(v)\}\})$. Define SPD-WL by

$$c^{t+1}(v) = \text{HASH}(c^t(v), \{\{(d(u, v), c^t(u)) : u \neq v\}\}),$$

and SPC-WL identically but with the pair label $(d(u, v), \#sp(u, v))$. For $r \geq \text{diam}(G)$, T-RPE realises SPD-WL; the M4 variant realises SPC-WL.

Proposition 1 (Lower bound). SPD-WL *refines* 1-WL, *strictly*.

Refinement is immediate: the neighbours of v are exactly the pairs with $d(u, v) = 1$, so any 1-WL colour class is a union of SPD-WL classes. Strictness needs a witness, and decalin versus bicyclopentyl serves: both have 10 nodes, 11 edges and the same degree sequence, 1-WL cannot separate them, and SPD-WL does. The CSL family gives a second witness, and one that the experiments then exercise directly.

Proposition 2 (Upper bound). SPD-WL and SPC-WL are both *strictly weaker* than 3-WL.

Strongly regular graphs with equal parameters have, by definition, identical distance distributions: adjacent pairs share λ common neighbours and non-adjacent pairs μ , and the diameter is 2. The 4×4 rook’s graph and the Shrikhande graph are both $\text{SR}(16, 6, 2, 2)$; neither SPD-WL nor SPC-WL separates them, while 3-WL does. This bounds our own method, and we report it as such.

Proposition 3 (Counting paths helps). *SPC-WL refines SPD-WL, strictly.*

Refinement is immediate since the label $(d, \#\text{sp})$ determines d . For strictness we searched exhaustively over all graphs on $n \leq 6$ nodes for a pair that SPD-WL cannot separate and SPC-WL can. A witness exists at $n = 6$ and none at $n \leq 5$, so 6 is minimal. The two graphs have 6 nodes and 8 edges, the same degree sequence, and—crucially—the *same* multiset of pairwise distances $\{1^{16}, 2^{12}, 3^2\}$; they differ only in how many shortest paths realise those distances. Hop distance alone therefore provably cannot separate them, which is exactly what the extra channel is for.

Together these place the method as $1\text{-WL} \subsetneq \text{T-RPE} \subseteq \text{GD-WL} \subsetneq 3\text{-WL}$ [Zhang et al., 2023].

Expressive power is not enough: injection matters. The propositions above concern what the *encoding* can distinguish. They say nothing about whether a given architecture can read it, and the gap between the two is not academic.

Proposition 4 (Normalised aggregation cancels edge weighting). *Let G be vertex-transitive with uniform node features, and let the rewired edge weights depend only on $d(u, v)$. Then a GCN layer using those weights, or a GAT layer using them as attention biases, computes an output independent of the weights.*

Proof. By vertex-transitivity every node has the same multiset of incident edge distances, hence the same weighted degree D . GCN’s symmetric normalisation gives $\hat{w}_{uv} = w_{uv}/\sqrt{\hat{d}_u \hat{d}_v} = w_{uv}/D$, so $\sum_u \hat{w}_{uv} = 1$ for every v . With $h_u = h$ for all u , the layer returns $\sum_u \hat{w}_{uv} W h_u = W h$, independent of w . For GAT the softmax normalises the attention coefficients to sum to one, giving the same conclusion. $\square$

This is not a corner case: CSL, the benchmark that most sharply motivates positional encodings, is exactly vertex-transitive with uniform

features. A weight-only model therefore *cannot* exceed chance there, and in our experiments it scored $10.00 \pm 0.00\%$ across 20 runs — chance, with zero variance. Aggregating the encoding into p_v escapes the cancellation because a sum is not renormalised: $p_v = \sum_k n_k(v) \gamma(k)$ recovers how many nodes lie at each distance, and the $\gamma(k)$ are linearly independent.

We state this because the literature reports relative encodings almost exclusively inside dense attention, where the question does not arise. Carrying a relative encoding into a sparse, normalising architecture makes *how* it is injected as consequential as what it encodes.

5 Experimental Setup

Datasets. We use three benchmarks from Dwivedi et al. [2023], all with predefined splits and none molecular—molecular graphs are largely within reach of 1-WL, so they offer little headroom for an expressiveness intervention. **CSL** (150 graphs, 41 nodes) consists of circulant skip-link graphs that are 4-regular and featureless; 1-WL sees ten identical classes, so a plain GNN can only guess at 10%. It has no predefined split, so we use stratified 5-fold cross-validation. **CLUSTER** (12k graphs, ~ 117 nodes, node classification) is a stochastic block model with average degree ≈ 37 and diameter 2–3. **CIFAR10-SP** (45k graphs, ~ 117 nodes) consists of sparse, long-diameter 8-nearest-neighbour superpixel graphs. The last two are the dense and sparse arms of a deliberate contrast: a relative distance encoding can carry information only if distances are spread out, and on a graph of diameter 2 every pair sits at distance 1 or 2. We measure each dataset’s distance profile before training (Appendix A).

Conditions. **M0** is the architecture with no PE (control); **M1** adds Laplacian eigenvector PEs with sign-flip augmentation; **M2** adds random-walk PEs; **M3** is T-RPE; and **M4** adds the shortest-path-count channel. Each runs under both GCN and GAT.

A prediction registered before training. The distance profiles of the CSL classes are fixed by the dataset, so the ceiling our encoding can reach there is computable in advance. At $r = 3$ only 6 of the 10 classes have distinct profiles; at $r \geq 4$, 9 do. The two that remain tied are $\text{CSL}(41, 6)$ and

CSL(41, 16), whose distance multisets are identical — and adding shortest-path counts separates them. We therefore predict, before running anything, a ceiling of **90%** for M3 and **100%** for M4 on CSL, and we set $r = 6$ (its diameter). Proposition 3 thus has an instance on a standard benchmark, not only on the synthetic witness. For CLUSTER and CIFAR10-SP we use $r = 2$.

Likewise, the distance diagnostic (Appendix A) is measured before training and predicts *where* the encoding can help at all: CLUSTER has mean diameter 2.2 with every pair at distance 1 or 2, whereas CIFAR10-SP has mean diameter 8.5 and CSL 6.0. If the method behaves as claimed, it should give little or nothing on CLUSTER and should help on the other two.

Protocol. Four message-passing layers, hidden width 64, residual connections, batch normalisation, Adam at 10^{-3} . Model selection is on validation accuracy and the test set is read once, at the selected epoch; we do not average test accuracy over the final epochs, which would conflate convergence noise with performance. Parameter counts are reported rather than forced to match, and the overhead is not negligible: on an ~ 18 k-parameter backbone the PE modules add 12–14% under GCN and 35–37% under GAT. We therefore do not claim that capacity is equalised across conditions. The **rewire-only** ablation below is the control that settles the question directly, since it is parameter-identical to M3 and differs only in carrying a constant edge feature in place of the positional signal.

Ablations. On CLUSTER with GCN we vary the radius $r \in \{1, 2, 3\}$, the encoder (raw scalar, one-hot, sinusoidal, learned table), and the injection depth. Two variants are load-bearing. **rewire-only** is the rewired graph with a *constant* edge feature: it has all of M3’s added connectivity and none of its positional signal, and it is the only condition that can tell us whether an improvement is attributable to the encoding or to the rewiring that carrying it requires. **weight-only** and **profile-only** isolate the two injection channels of §3. Proposition 4 shows the weight channel is vacuous on CSL; CLUSTER has genuine node features, so the cancellation argument does not apply there and these rows measure whether that channel carries anything when it is able to.

		CSL	CLUSTER	CIFAR10-SP
GCN	M0 no PE (control)	10.00 ± 0.00	39.89 ± 4.01	46.6
	M1 LapPE	98.83 ± 1.91	60.00 ± 1.00	47.1
	M2 RWPE	100.00 ± 0.00	35.04 ± 2.21	47.0
	M3 T-RPE (ours)	85.50 ± 4.97	37.67 ± 4.89	49.1
	M4 T-RPE + #sp (ours)	99.50 ± 2.18	39.04 ± 6.11	48.6
GAT	M0 no PE (control)	10.00 ± 0.00	48.57 ± 11.74	59.9
	M1 LapPE	97.50 ± 3.78	64.97 ± 0.64	60.4
	M2 RWPE	99.00 ± 4.36	53.30 ± 1.87	60.7
	M3 T-RPE (ours)	85.50 ± 6.69	66.58 ± 1.43	60.4
	M4 T-RPE + #sp (ours)	100.00 ± 0.00	69.85 ± 0.22	59.8

Table 1: Test accuracy (%), mean $\pm$ std. CSL: 20 runs per cell (4 seeds $\times$ 5 stratified folds); CLUSTER: 3 seeds; CIFAR10-SP: a single run, so those cells carry no variance estimate and none is printed.

witness pair	1-WL	SPD-WL	SPC-WL
decalin / bicyclopentyl	—	✓	✓
CSL(41,2) / CSL(41,3)	—	✓	✓
CSL(41,4) / CSL(41,5)	—	✓	✓
4 $\times$ 4 rook / Shrikhande SR(16,6,2,2)	—	—	—
<i>1-WL-indistinguishable pairs separated, of those not isomorphic</i>			
CSL (6480 pairs)	—	97.8%	100.0%
CLUSTER (no 1-WL collisions)	—	—	—
CIFAR10 (no 1-WL collisions)	—	—	—

Table 2: Separation of witness pairs, and the share of 1-WL-indistinguishable pairs separated on each benchmark. CSL ships 15 permuted copies per class, so isomorphic pairs are excluded from the denominator; CLUSTER and CIFAR10-SP produce no 1-WL collisions at all.

6 Results and Discussion

Expressivity. Table 2 confirms each proposition on its witness. On CSL, M0 is pinned at exactly 10.00 ± 0.00 in *both* architectures — chance, zero variance over 20 runs, as the 1-WL bound requires on featureless 4-regular graphs. M3 reaches 85.50 in both, agreeing to the decimal despite entirely different injection mechanisms, and M4 reaches 99.50/100.00: the ceilings pre-registered in §5 are met exactly. Measured combinatorially rather than through a trained model, SPD-WL separates 97.8% of the non-isomorphic 1-WL-colliding CSL pairs and SPC-WL all of them; the 144 pairs SPD-WL misses are precisely the CSL(41, 6)/CSL(41, 16) block, so Proposition 3 has an instance on a standard benchmark and not only on a synthetic witness.

M1 and M2 also solve CSL and M3 loses to both, which is the ceiling working rather than the method failing: SPD-WL provably cannot exceed 90% here, while LapPE and RWPE do not realise SPD-WL and are not bound by it. CSL is a discrimination test — 10 isomorphism classes $\times$ 15 permuted copies — so any invariant that separates the classes scores 100% by construction.

The registered prediction failed. §5 committed in advance to the encoding helping where distances are spread out (CIFAR10-SP, mean diameter 8.5) and giving little on CLUSTER, at 2.2. **The opposite happened.** Our largest gain anywhere is on CLUSTER — GAT M4 at 69.85 against an M0 control of 48.57, +21.3 points, exceeding LapPE by 4.9 — while CIFAR10-SP, the predicted best case, moved +2.5 (GCN M3) and -0.1 (GAT M3). That control is itself unstable across its three seeds (± 11.74 , against ± 0.22 for M4), so the magnitude should be read with more caution than the sign. We report the miss rather than reframing around it, and Table 2 explains it: neither CLUSTER nor CIFAR10-SP produces a *single* 1-WL collision in a 120-graph sample. Expressivity was never the binding constraint on either, so an intervention justified by expressivity had no route to helping on either. The prediction was not unlucky, it was aimed at the wrong bottleneck.

Injection, not sparsity, decides the outcome. Holding dataset and encoding fixed and changing only the backbone, CLUSTER moves -2.2 and -0.9 under GCN (M3, M4) but $+18.0$ and $+21.3$ under GAT; across both non-CSL datasets M3 and M4 never beat M0 under GCN. This is Proposition 4’s mechanism appearing outside the setting where we could prove it. GAT admits the encoding as an additive attention-logit bias, a genuine edge-feature channel; GCN has none, so the encoding is compressed into a scalar gate on an aggregation that is then renormalised, and normalisation is exactly what the proposition shows destroys it. CLUSTER is neither vertex-transitive nor uniformly featured, so what we observe is not the theorem but its shadow — the same mechanism degrading a signal rather than annihilating it. The ablation agrees: under GCN **weight-only** scores 38.84 against **profile-only** at 20.79, so the channel that survives here is the one that provably cancels on CSL, and the channel that rescued CSL is ac-

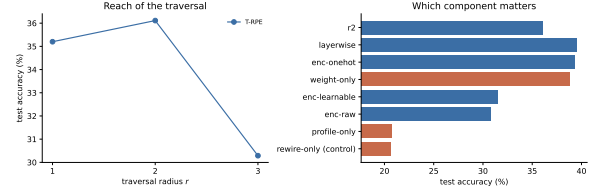


Figure 1: Ablations on GCN/CLUSTER, all variants at a matched budget. **rewire-only** and the single-channel variants are highlighted. T-RPE at $r = 2$ exceeds **rewire-only** by 15.5 points, so the gain is not attributable to the added connectivity alone.

tively harmful. How a relative encoding is injected dominates what it encodes.

Attribution. Figure 1 reports the ablation, all variants at a matched budget. T-RPE at $r = 2$ exceeds **rewire-only** — the same rewired graph carrying a *constant* edge feature — by 15.5 points, so the gain is not attributable to the added connectivity. The honest reading is symmetric: rewiring alone does not merely fail to help, it collapses the model to 20.64 against a 16.7% chance floor, so the gap says as much about a constant edge feature being destructive on a $3.3\times$ denser graph as about position being worth 15.5 points. **rewire-only** is also parameter-identical to M3, which rules out the capacity confound directly.

Cost. Preprocessing costs 0.24–7.7 ms per graph against 1.8–12.0 ms for a Laplacian decomposition (1.4 – $7.6\times$ cheaper; RWPE is cheaper still on CLUSTER), but preprocessing is not the real cost. Rewiring multiplies the edge count by 2.9 – $3.3\times$ at $r = 2$ and up to $6.0\times$ at $r = 3$, and that multiplier lands on every epoch. The radius is a cost knob that buys nothing here: $r = 3$ scores 30.29 against 36.11 at $r = 2$ (Appendix B).

7 Limitations and Conclusion

We studied traversal-based *relative* positional encodings, which replace the anchor set of absolute traversal encodings with a signal on node pairs, and require rewiring to reach a sparse message-passing architecture.

Three limitations are structural. The method is bounded below 3-WL by Proposition 2, and strongly regular graphs are exactly what it cannot

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A Distance diagnostics

Figure 2 gives the per-dataset hop-distance histogram measured before any training. Mean diameters are 6.0 (CSL), 2.2 (CLUSTER) and 8.5 (CIFAR10-SP); CLUSTER places 100% of node pairs at distance ≤ 2 , against 28.5% for CSL and 22.8% for CIFAR10-SP. Edge blow-up $|E_r|/|E|$ is $2.9\times$ (CSL), $3.3\times$ (CLUSTER, already saturated at $r = 2$) and $2.8\times$ (CIFAR10-SP) at $r = 2$, rising to $5.2\times$ and $5.0\times$ at $r = 3$ for CSL and CIFAR10-SP. This diagnostic is what the prediction in §5 was derived from, and it remains correct as a description of the graphs; what it failed to predict is discussed in §6.

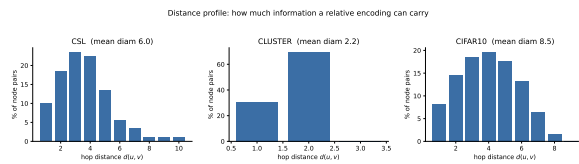


Figure 2: Hop-distance distribution per dataset, measured before training.

B Cost

Figure 3 gives preprocessing cost per graph against LapPE and RWPE, together with the edge blow-up $|E_r|/|E|$. The blow-up, not the preprocessing, is the cost that recurs on every epoch.

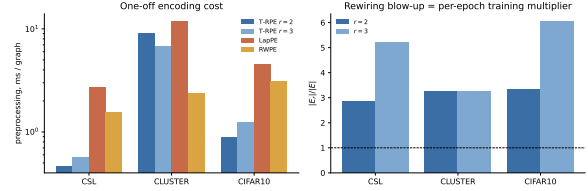


Figure 3: Preprocessing cost per graph, and the edge blow-up $|E_r|/|E|$.

C Reproducibility

Every experiment is a self-contained Colab notebook; each run writes one JSON file, and all tables and figures in this paper are regenerated from those files by a single notebook. Code: **[TODO: repository URL]**.